

FALCON's Response to Delhivery for RFQ of Sorter 10K_Coimbatore

April 8, 2024





Kind Attention –

Mr. Sunny Raja

Mr. Himanshu Shekhar

M/s DELHIVERY

Offer Ref: F24-00175-02; Date 08.04.2024.

Subject – Techno -Commercial Offer of Shipment Sorter

We are pleased to submit our Techno-Commercial Offer to DELHIVERY in response to your **RFQ for Shipment Sorter, Coimbatore.**

We would like to highlight the fact that Falcon Autotech, as a prime contractor for this project, is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative. From technical point of view, the system is **designed at 10,000 PPH** and ensures simple operations and movement within the facility.

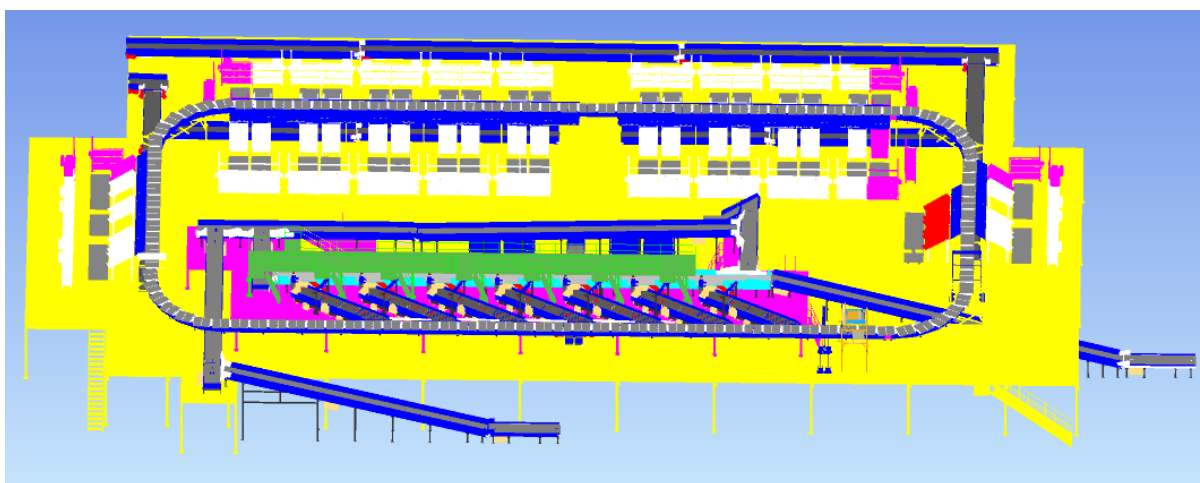
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFQ for Shipment Sorter_ Delhivery, Coimbatore



Proposal Reference – F24-00175-02

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Contents

Contents	4
1. Glossary	7
2. Executive Summary	8
3. Handled Parcel Spectrum	9
3.1 Parcel size loadable on the sorter	9
3.2 Parcels to be loaded on Sorter shall have the following characteristics:	9
3.3 Parcel not loadable on the sorter	10
3.4 Pictures of Sortable Parcels	11
4. Proposed System Description	12
4.1 Objective	12
4.2 Summary of the System	12
4.3 Main benefits of the proposed solution	13
5. Layout View	14
6. Proposed System Capacity Calculations	15
6.1 Sorter System Capacity	15
7. System description	15
7.1 Infeed System	15
7.1.1 Infeed Conveyor System	16
7.1.1.1 Straight and Inclined Powered Belt Conveyor	16
7.1.2 VDS System:	18
7.1.2.1 VDS Loop Conveyor:	18
7.1.2.2 VDS Modular Conveyor	19
7.1.3 Conveyor BOQ	20
7.2 Feedlines	21
7.2.1.1 Orientation Conveyor	21
7.2.1.2 Weighing Conveyor	22
7.2.1.3 Buffer Conveyors	22
7.2.1.4 Intelligent Merge Conveyor	23
7.2.1.5 Irregular Chute	23
7.3 Irregular Takeout Conveyor	24
7.4 Main Loop	25
7.5 Sorting Output	26
7.5.1 PTL Chute	26
7.5.2 Rejection Chute	27
7.5.3 Collection Bin	28

8.1	Mechanical equipment	29
8.2	Electrical Equipment	30
9.	Description of Components of Equipment	31
9.1	Cross Belt Sorter -	31
9.2	CBS Carrier	31
9.2.1	Servo Roller.....	32
9.2.2	Chassis.....	32
9.2.3	Carrier Wheels	32
9.2.4	Friction Wheel Drive.....	33
9.2.5	Non-contact based linear motor drive	33
9.2.6	Power Transmission	33
9.2.7	Data Transmission	34
9.2.8	Carriers positioning System.	34
9.2.9	Technical specifications of Sorter	35
9.3	Scanning & Sensing on Main Sorter Loop.....	36
9.3.1	Automatic Barcode Scanning & Dimensioning System.....	36
9.3.2	Parcel Positioning System.	36
9.3.3	Empty Carrier Detection System.....	36
10.	Electrical System.....	37
10.1.1	Reference Picture of Power Distribution Panel	37
10.1.2	Main Control Panel (Reference).....	37
10.1.3	Induct Stations Control Panel (Reference).....	38
11.	Controls Architecture (Ref.)	39
12.	IT Architecture (Ref.)	39
13.	Key Components Make	40
14.	Program Organisation	41
14.1	Program/ Project Schedule	41
14.2	Program Management.....	41
15.	DELHIVERY'S Responsibility	41
15.1	DELHIVERY Responsibilities During the Assembly and Commissioning Phase	41
15.2	Responsibilities of DELHIVERY During the Tests	42
15.3	DELHIVERY Responsibilities During the Training.....	42
16.	Warranty Period	43
17.	Commercial Terms.....	44
17.1	Price Sheet	44
17.2	Payment Terms.....	44



18.	Exclusions	45
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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	ARB	Actuated Roller Balls
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this RFQ of Sorter. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.

As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the Sorter system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **One loop of sorter, based on a cross belt technology, offers a designed throughput of 10000 pph.**
- **Infeed conveyor system.**
- **6 Nos of Feedline (1 No is planned for future scope)**
- **45 PTL Chutes, 3 rejection chutes, and 6 Irregular chute.**

A tailor-made and simple layout, specifically designed to Delhivery The proposed layout, is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our last Teams workshop meeting on 2nd April 2024.

- Simple operational conditions due to one single loop cross belt sorter.
- Easy maintenance: optimized number of conveyors and concentrated inducts area.
- Vehicle movement zones.
- Exceptions handling area with direct feeding via conveyor.

1. **Falcon's reliable and fully CE compliant Parcel sortation systems**

These systems are globally being used by most innovative brands such as Aramex, Amazon, Asendia, Fastway, Delhivery and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Cognex and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the Delhivery Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from the completion of installation & commissioning.

3. **Commitment to ensuring the quality of the delivery and its schedule.**

Falcon proposes a ramp-up approach, with a soft go-live by End of Oct 2024.

3. Handled Parcel Spectrum

As per the parcel spectrum data provided in the RFQ documents, Falcon has studied and analysed the parcel spectrum in detailed.

3.1 Parcel size loadable on the sorter

Falcon's heavy duty cross belt sorter has a capability to handle the below mentioned parcel sizes and weight.

Specification	Unit	Value
Max Length	mm	500
Max Width	mm	500
Max Height	mm	500
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	50
Min Height	mm	5
Min Weight	Kg	0.05
Dimension Accuracy	mm	+/- 10
Weighing Accuracy	Kg	+/- 60 grams or 0.05% of the Weight of Product (whichever is higher)

Note- Dimension will only be captured for parcels > 20 mm in height .

3.2 Parcels to be loaded on Sorter shall have the following characteristics:

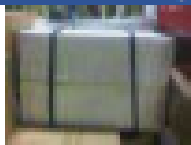
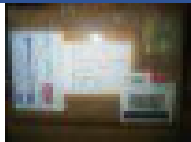
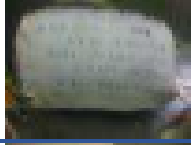
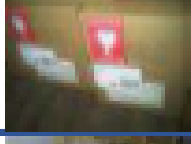
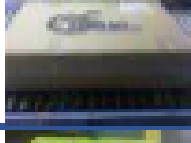
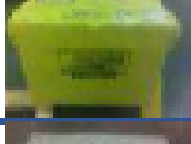
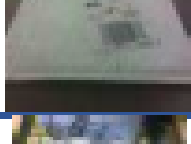
- Centre of Gravity of item must not move during conveyance or sorting.
- Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
- Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
- Plastic ropes shall be perfectly adherent to the surface of the package.
- All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
- Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
- Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
- The parcels must have at least one flat and regular surface providing enough stability during conveyance.
- All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
- All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

3.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

3.4 Pictures of Sortable Parcels

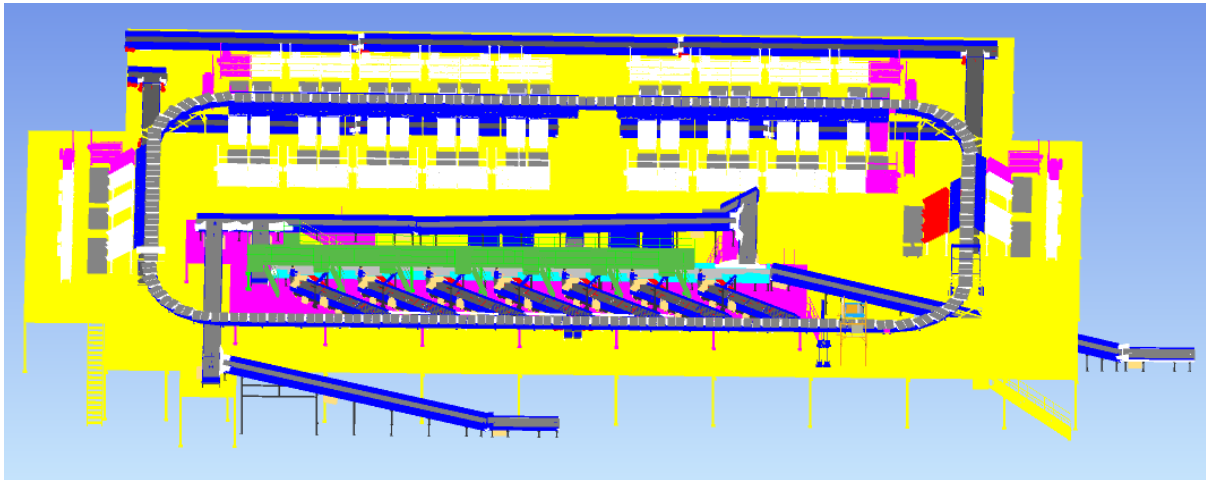
PARCEL TYPE	DESCRIPTION
	Strongly Strapped A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item A cardboard parcel coated in a packing paper.
	Cloth Covered Item Often import parcels that are of irregular shape.
	Wine Box A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box A cardboard parcel is often the norm.
	Totes A large plastic, lidded box.
	Flats A parcel that often contains documents.
	Poly Bags It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

4. Proposed System Description

4.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Loop CBS for sorting shipments into destination as per specifications shared by DELHIVERY.

4.2 Summary of the System



The complete parcel processing system includes, in accordance with the specifications provided by **DELHIVERY**:

Process Flow:

- Shipments will be loaded on the Infeed Conveyor placed on ground level
- Straight PVC Conveyor and Inclined Conveyor will move the shipments to the three VDS Loop System.
- From the VDS Modular Conveyor , Operator will pickup the parcels and placed it on the induct line.
- Operators orients and places the shipments on the Induct lines equipped with sensors
- Feedlines induct the parcels on the Sorter automatically onto to the Cross-Belt Sorter Loop which then sorts the parcels into respective chute using data from Delivery's Sorting decision system. The Dimensions and image of the parcel is also captured on the Sorter.

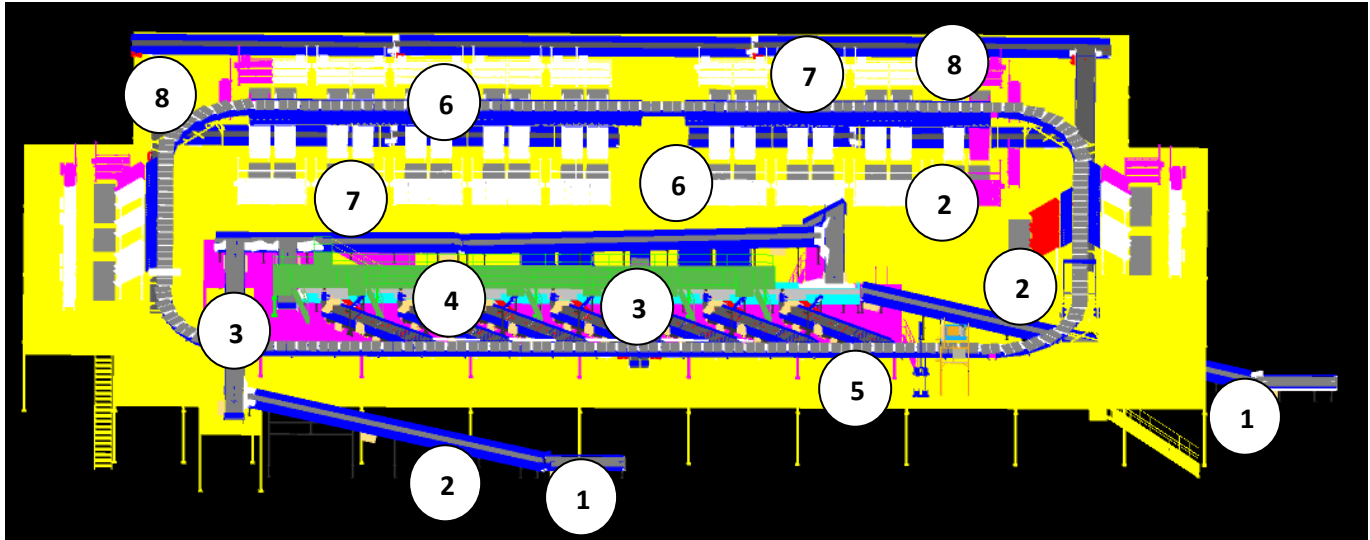
- The sorted shipments are discharged into PTL Chutes.
- Secondary Sorting happens at PTL Chutes with help of Put to Light Racks and Bags are prepared. Associates scan the Shipments and drops them into the Bags guided by Light Modules on Racks.
- Take out Conveyors are planned near the chutes. Once the Bags are filled, they are dropped onto the Take-out Conveyors.
- Bag Takeout conveyors will transfer the bags to the spiral chute which is in client scope

4.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 2.1 m/s.
3. Low occupancy of floor space in the building
4. Narrow discharge centers for increased number of splits in limited space
5. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

5. Layout View



Legend

- 1 – Infeed Conveyor
- 2 – Infeed Inclined Conveyors
- 3 – VDS Loop Conveyor
- 4 – Feedlines
- 5- Main Loop
- 6- PTL Chute
- 7- PTL Frames
- 8- Bag takeout Conveyor

6. Proposed System Capacity Calculations

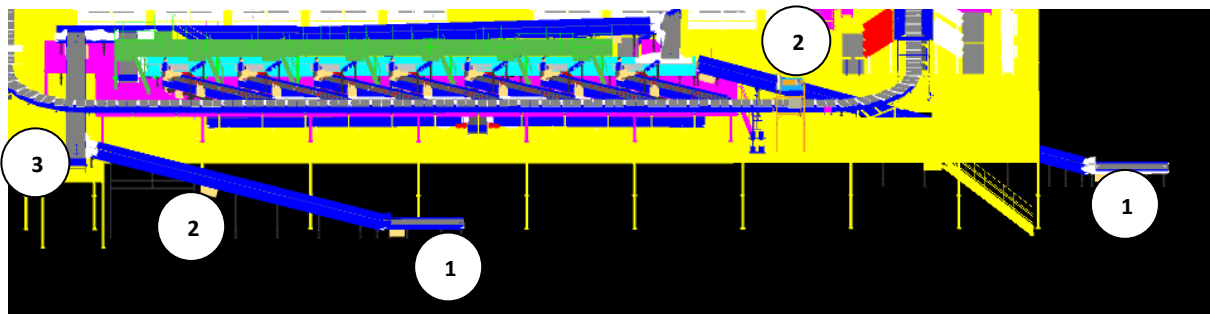
6.1 Sorter System Capacity

The following table shows the throughput calculation for the sortation system designed based on DELHIVERY's RFP requirements.

SPECIFICATION	VALUE
Sorter Type	Loop CBS
Speed	2m/s
Pitch	1,175 mm
Carrier per hour	6,127 CPH
Belts per hour	12,254 BPH
Sorter Designed Capacity	12254 PPH per deck
Feedline Design Capacity	1800 PPH
No of Inducts Required for 10K PPH	6 Nos

7. System description

7.1 Infeed System



Legend

- 1- Infeed Straight Conveyor
- 2- Infeed Inclined Conveyor system
- 3- Infeed Highway Conveyor

7.1.1 Infeed Conveyor System

Infeed conveyor system consists of below components:

1. Inclined PVC Conveyor
2. Straight PVC Conveyor
3. Curve Conveyor

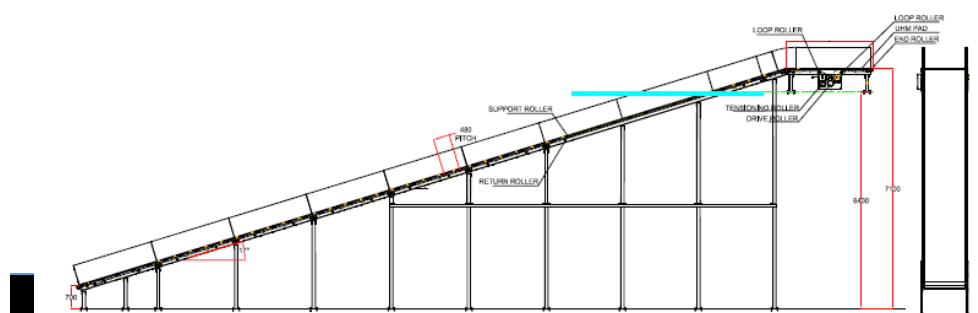
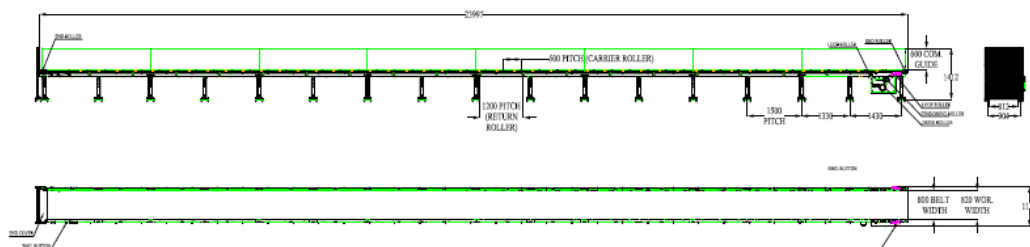
7.1.1.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

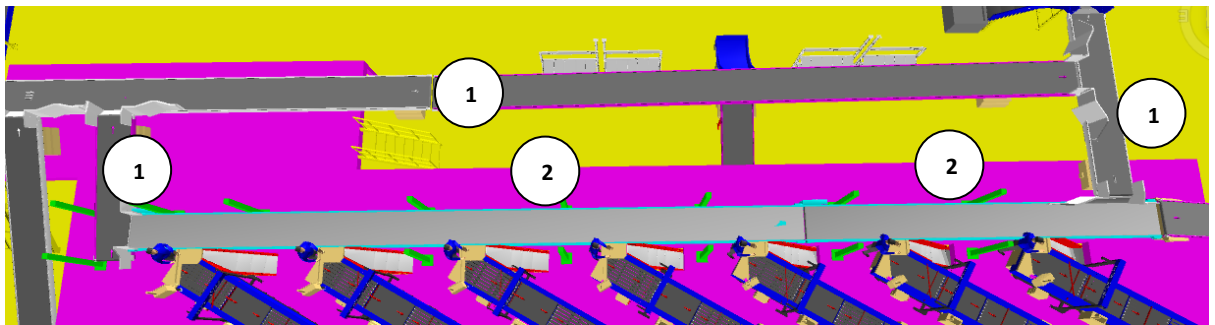


Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000/1200/1400
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

7.1.2 VDS System:

Volume distribution system is designed to manage and distribute shipments efficiently along the feedlines. Volume distribution System has below key 18 component:

- VDS Loop Conveyor
- VDS Modular Conveyor



Legend

1. VDS Loop Conveyor
2. VDS Modular Conveyor

7.1.2.1 VDS Loop Conveyor:

Bulk materials will enter in to the VDS Loop through Infeed Highway Conveyors. Then the material will get distributed in to different VDS Chute maintaining load balance. The remaining material gets circulated in the VDS loop till the VDS Chute not be available for sorting .



7.1.2.2 VDS Modular Conveyor

In this solution we have designed modular conveyor as VDS Conveyor . Operator will pick the parcels from this modular conveyor



7.1.3 Conveyor BOQ

S No.	Name	ANGLE	EL_1	EL_2	Conveyor Length (m)	Conveyor width (mm)	Set
1	Modular Conveyor	0	7050	7050	20160	20.16	1
2	Modular Conveyor	0	7050	7050	10620	10.62	1
3	PVC Belt Conveyor with Torque Arm	0	4850	4850	2400	2.4	1
4	PVC Belt Conveyor	0	7050	7050	13440	13.44	1
5	PVC Belt Conveyor	1.7	6950	7550	19680	19.68	1
6	PVC Belt Conveyor	0	950	950	4320	4.32	1
7	PVC Belt Conveyor	15.2	4750	7300	9600	9.6	1
8	PVC Belt Conveyor	15.4	600	5100	16800	16.8	1
9	PVC Belt Conveyor	0	700	700	4320	4.32	1
10	PVC Belt Conveyor	16.1	850	7150	22560	22.56	1
11	PVC Belt Conveyor	9	6600	7300	4320	4.32	1
12	PVC Belt Conveyor	0	7300	7300	5760	5.76	1
13	Bagging Conveyor	-0.4	4600	4700	14400	14.4	1
14	Bagging Conveyor	-0.5	4600	4700	12000	12	1
15	PVC Belt Conveyor	9.9	4600	5450	4800	4.8	1
16	Bagging Conveyor	-1.5	4600	4700	3600	3.6	1
17	Bagging Conveyor	-1.4	4600	4700	5281	5.281	1
18	Bagging Conveyor	-0.5	4600	4700	12000	12	1
19	Bagging Conveyor	-0.5	4600	4700	12000	12	1
20	Bagging Conveyor	-0.5	4600	4700	12480	12.48	1
21	Bagging Conveyor	-0.3	4600	4700	19680	19.68	1
22	Bagging Conveyor	-0.4	4600	4700	13920	13.92	1
23	Bagging Conveyor	-2.2	4600	4700	2402	2.402	1
24	Bagging Conveyor	-0.4	4600	4700	14400	14.4	1
25	Bagging Conveyor	-0.9	4600	4700	6241	6.241	1
26	Bagging Conveyor	-0.3	4600	4700	19680	19.68	1
27	Bagging Conveyor	-0.6	4600	4700	9121	9.121	1
28	Bagging Conveyor	-0.5	4600	4700	12480	12.48	1

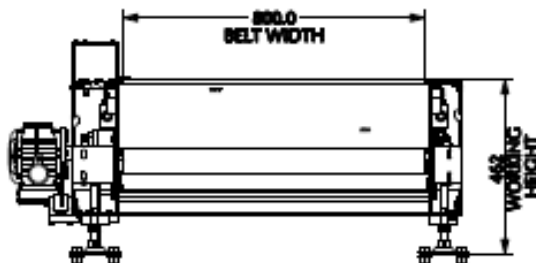
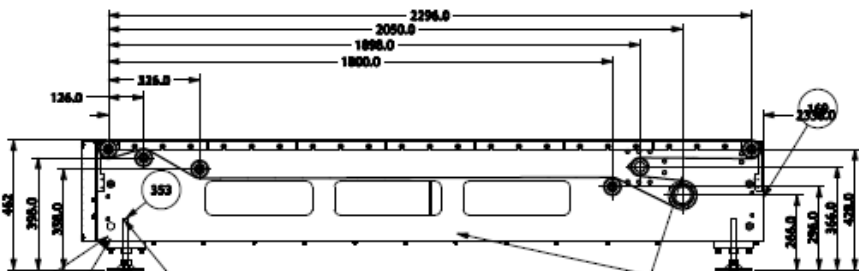
7.2 Feedlines

Loop CBS has a total 6 Automatic Feedlines (1 No is for future expansion) with a Designed IPP of 2000 PPH.

1. Each Feedline has Four Conveyor Modules-
 - Orientation Conveyor- 1 Nos.
 - Weighing Conveyor- 1 Nos.
 - Buffer conveyor- 1 Nos.
 - Intelligent Merge- 1 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
3. Over the feedline, shipments are evenly spaced and smoothly transferred to the main Loop.

7.2.1.1 Orientation Conveyor

An Orientation conveyor is a type of conveyor system used to align & Release the products for induction onto CBS. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.



7.2.1.2 Weighing Conveyor

A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process. Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

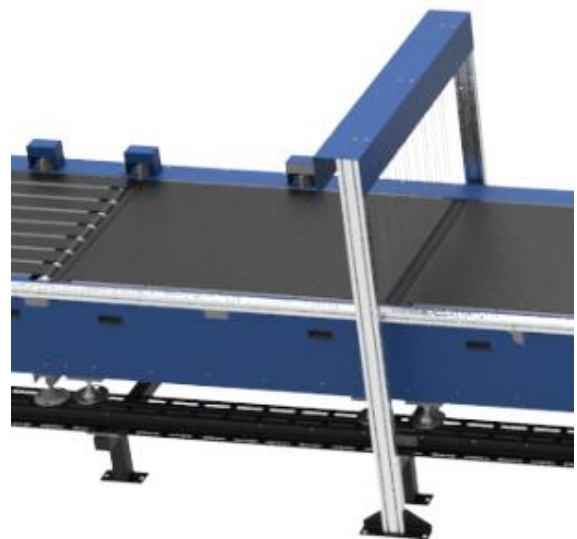
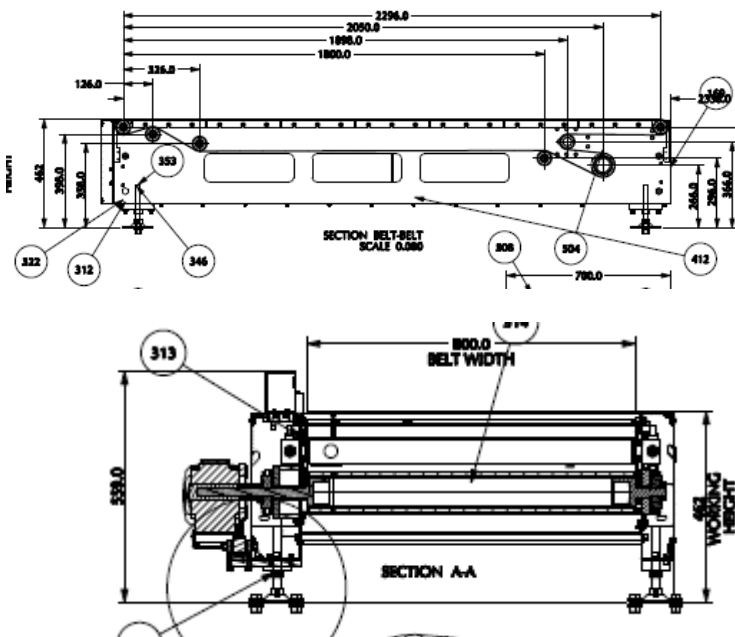
Make- Bizerba/ Mettler Toledo



7.2.1.3 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

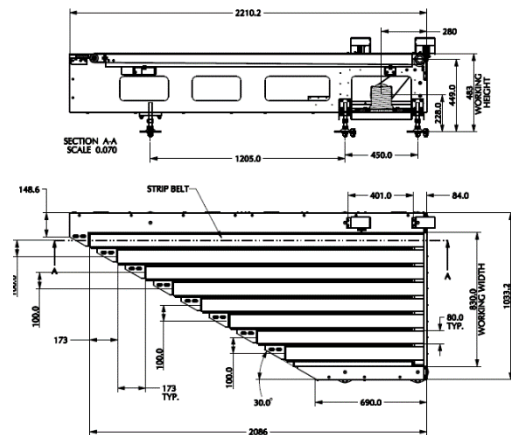
This Conveyors required to maintain the Throughput of Line.



7.2.1.4 Intelligent Merge Conveyor

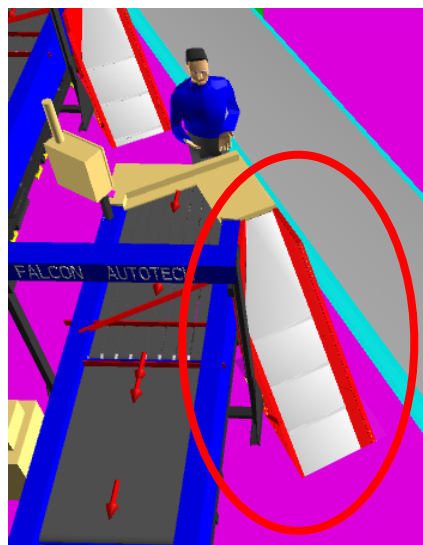
An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular high-speed conveyor used for inducing shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



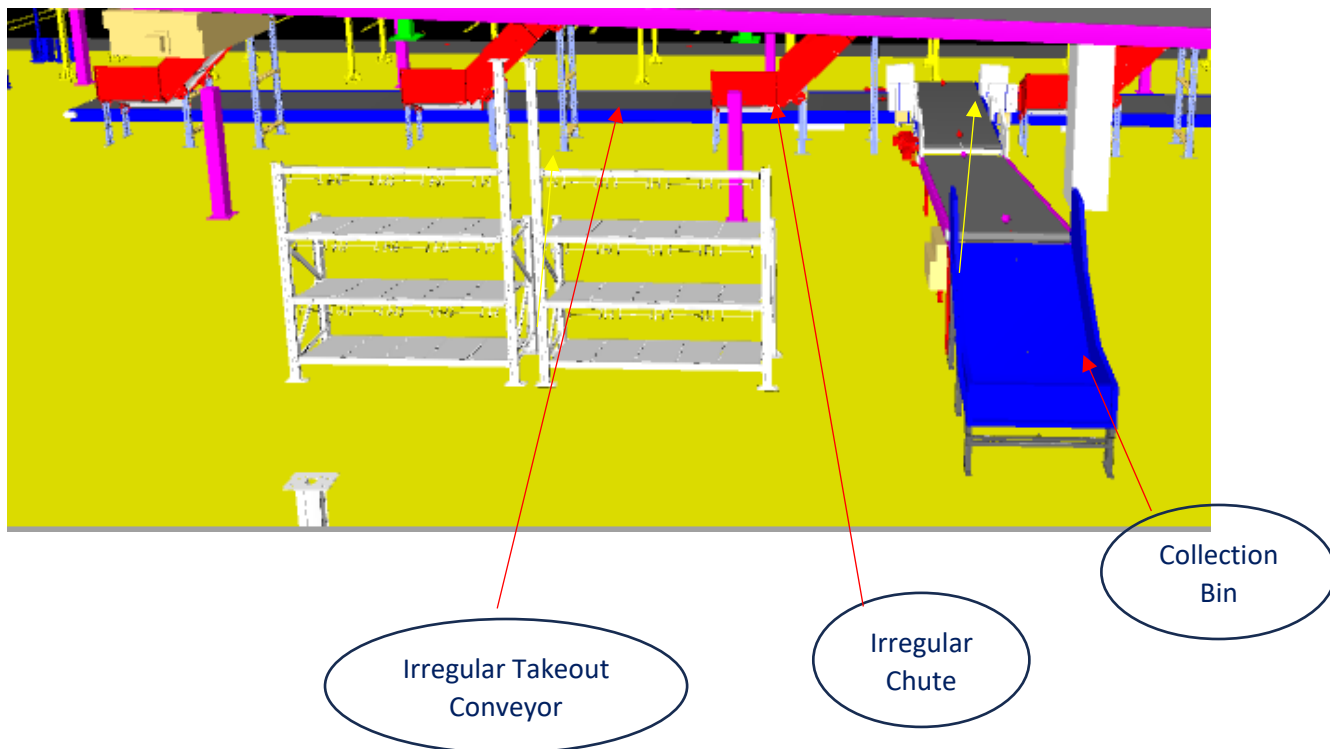
7.2.1.5 Irregular Chute

On each Feedline one irregular chute is planned. Any shipments which is not inductable on Sorter will be drop into this chute manually by Operators. Shipment will drop on irregular takeout conveyor via these chutes.

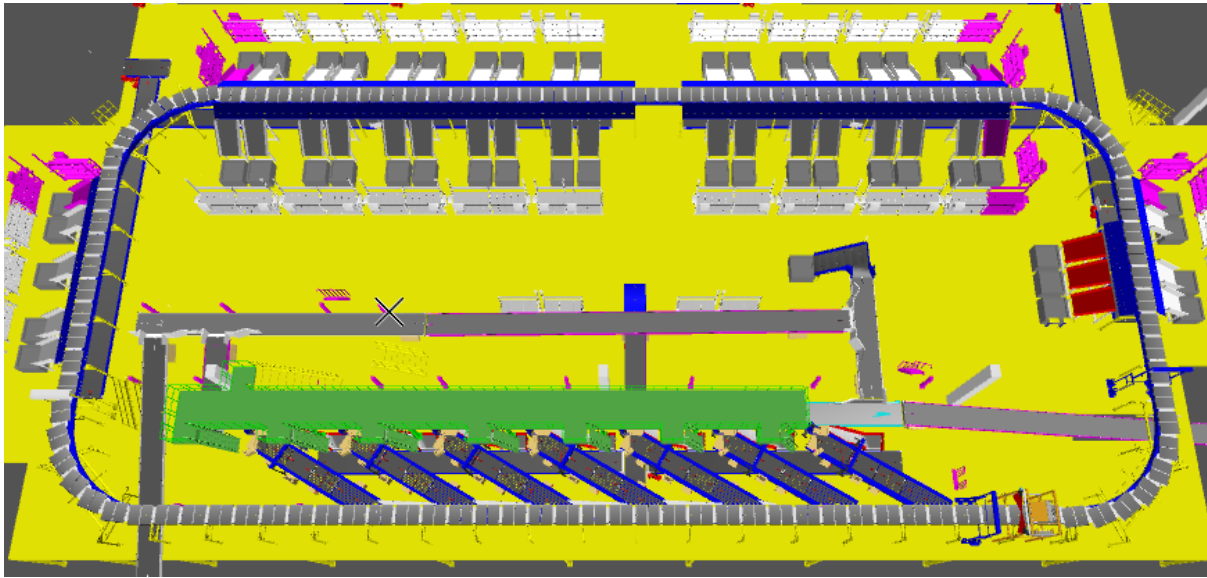


7.3 Irregular Takeout Conveyor

One conveyor for each induct zone is planned to takeout the irregular shipments. This conveyor will collect the shipments from irregular chutes and drop them in a bin for manual process.



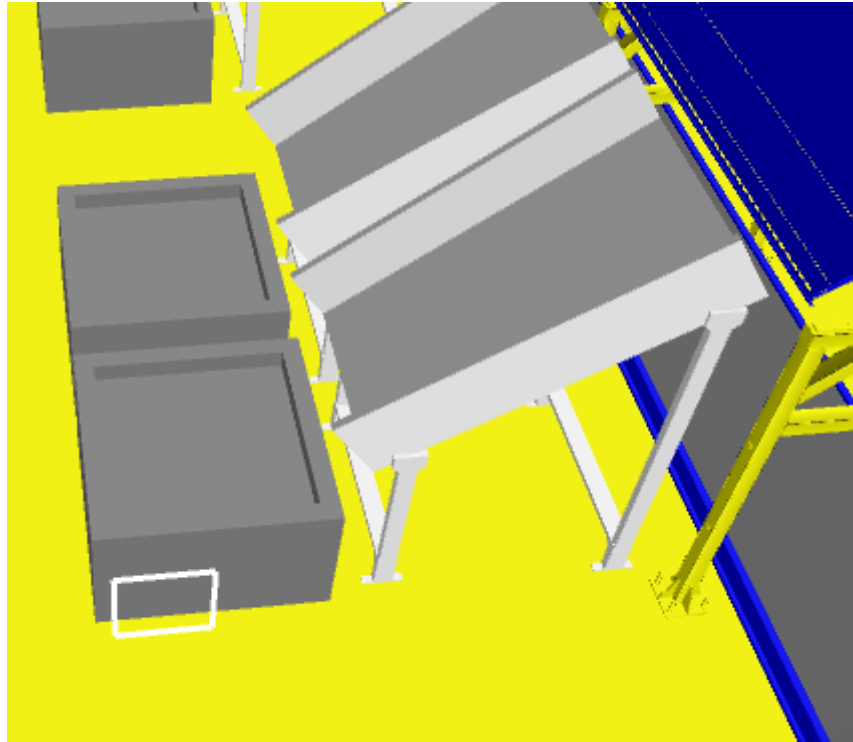
7.4 Main Loop



1. The proposed CBS Sorter will be installed on mezzanine. The Sorter runs at 2500 mm from mezzanine level .
2. We have provided loop with dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.
3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

7.5 Sorting Output

7.5.1 PTL Chute



- These are collection type chutes where the parcels are collected into chutes.
- The Operators collects the parcels from chute and will do secondary sorting by means of PTL Racks.

Chute IOs	Description
Single Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Tower Lamp	To indicate the operator action on a chute
Push Button	To Suspend/Resume Sorting into a particular Chute

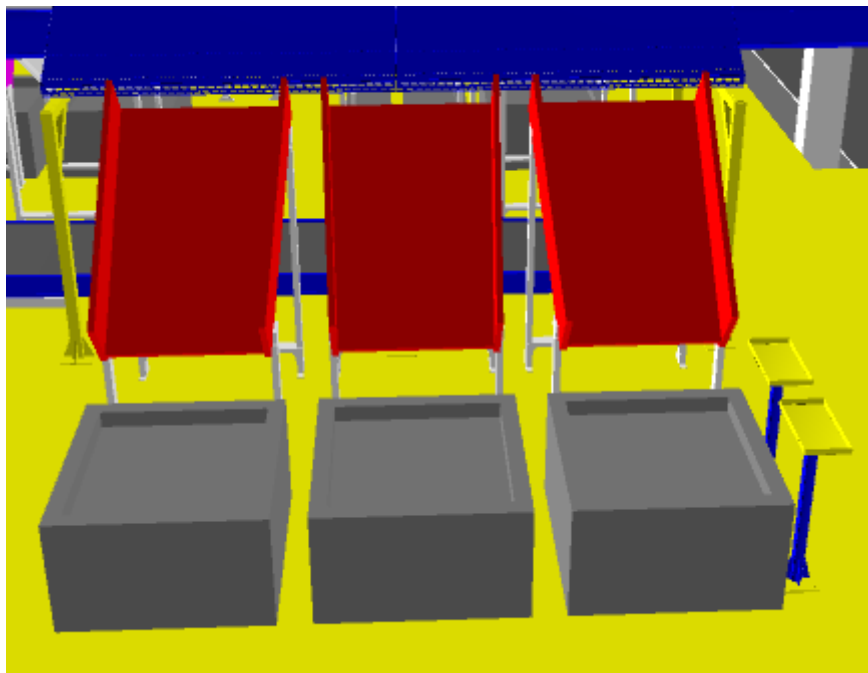
PTL Racks & Modules

- PTL Racks with PTL light modules has been provided for secondary sorting of parcels that have been dropped in the PTL chutes.
- Rack Configuration: 3 Rows, 4 Columns, total of 12 modules/rack in one chute.



- Operators will pick up a parcel from the PTL chute and scan the parcel barcode by handheld scanner.
- The light of the block in the PTL rack which has been mapped to the destination location of the parcel will glow.
- Operator will place the parcel in the bag framed in that module and acknowledge by pressing the confirmation button on the module.
- This process will be repeated for all parcels falling in that chute bin.
- Once a bag is full, the operator will take out the bag paste specific tag on bag seal and place it on the bag take-away conveyor.
- The new bag will be placed in the empty module.

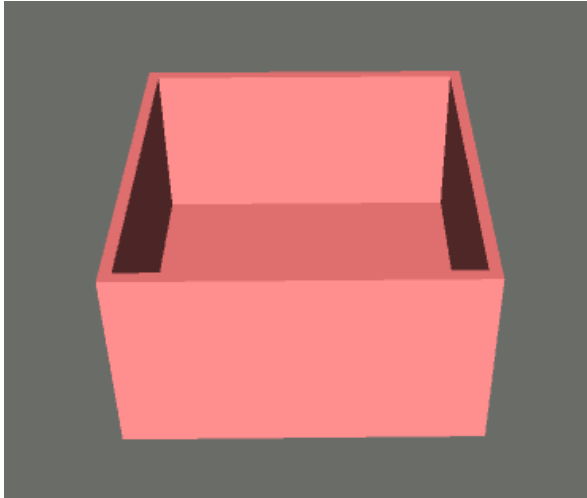
7.5.2 Rejection Chute



System is installed with 3 Nos rejection chutes Once the parcel got rejected, it will stage on the Collection Bins for the manual process.

7.5.3 Collection Bin

Spring Loaded Bin is planned on each chute to collect the parcels.



8. Proposed System Technical Details

8.1 Mechanical equipment

Pos.	Qty.	Description	Value
FM1	1	Infeed System - Powered Belt Conveyors - Modular Belt Conveyor	~278 Metres (26 Conveyors) ~31 Metres (2 Conveyors)
FM2	6	Feedlines Consists of - Loading Conveyor. - Weighing System - Dynamic Spacing System - Angle merge 1-D & 2-D Hand Scanners for Manual Scanning	1 Set 1 Set 1 Set 1 Set 1 No.
FM3	1	Falcon Sorter 1 Loop Cross Belt Sorter - Sorter Height - Sorter Length - Sorter Speed - Sorter Drive - Carrier Pitch Including: - Standard Sorter Supports - Empty Carrier Detection System - Product Centring System - Dimension & Top side Scanning System - E-Stops - Hooters - Fencing - Netting - Overshoot Assembly	2500 mm (from Mezzanine) ~135 m ~2 m/s Linear Motor 1175 mm
FM6	1	Chutes - PTL Chutes - Rejection Chute - Irregular Chute - Spring Loaded Fixed Bin	45 Nos 3 Nos 6 Nos 49
FM7	1	PTL - PTL Racks for Chute - PTL Racks for Manual* - PTL Modules 1-D & 2-D Hand scanner for PTL Racks - PTL Control Box	600 Nos 48 Nos 600 Nos 45 Nos 46 Nos



**Note: For Manual Racks, we didn't considered PTL and Hand scanner .*

8.2 Electrical Equipment

Electricals			
FE1	1	Consists of Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

9. Description of Components of Equipment

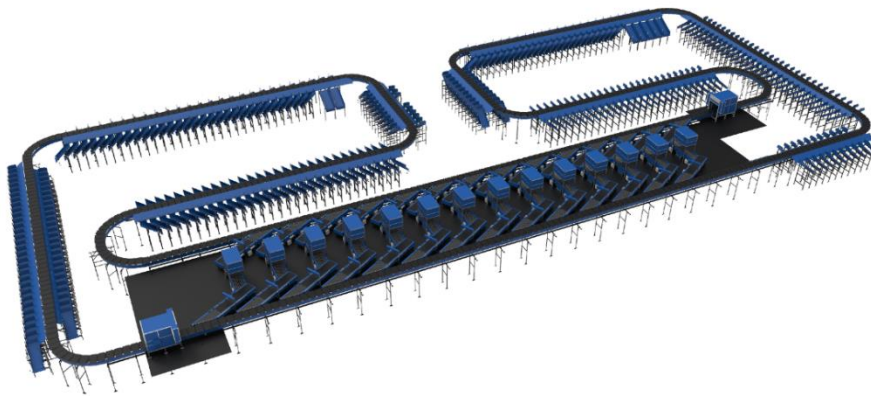
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

9.1 Cross Belt Sorter -

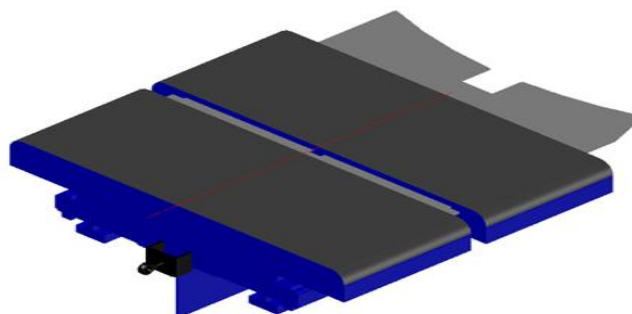
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



9.2 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners" and non-sortable parcel recirculation.



9.2.1 Servo Roller

High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.



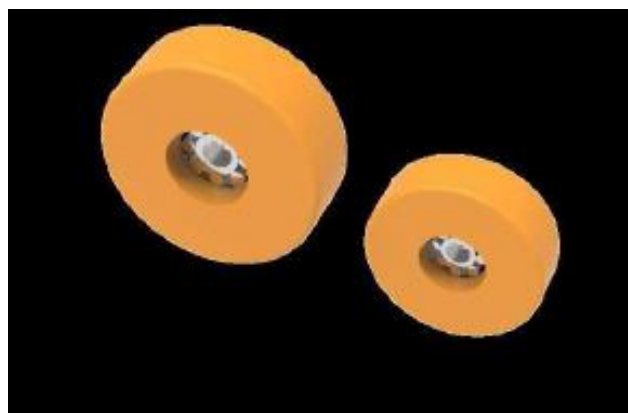
9.2.2 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.



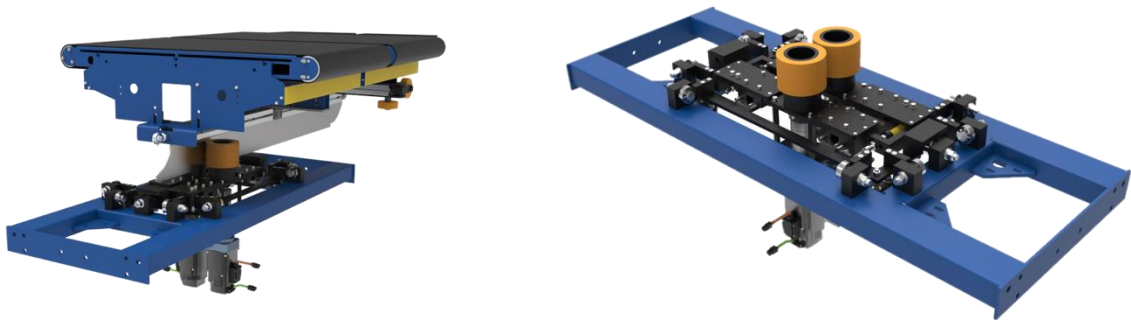
9.2.3 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle.



9.2.4 Friction Wheel Drive

Falcon can provide the indigenously developed Friction Wheel Drive (FWD) to drive the cross belt loop. This driving mechanism operates on the principle of friction, as the name implies. This unit comprises two independent motor-driven wheels that spin in opposite directions synchronised. FWDs are highly energy-efficient drives that promote **sustainability** compared to Linear induction/synchronous motors.



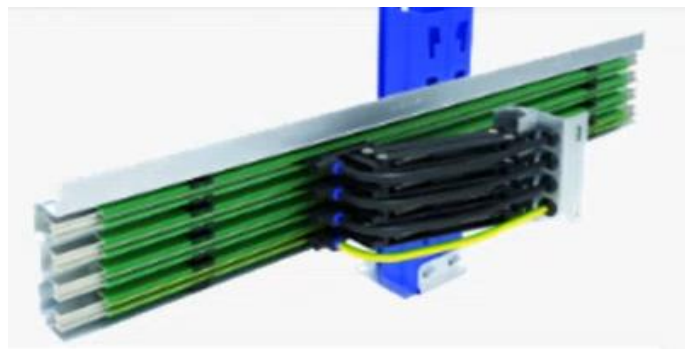
9.2.5 Non-contact based linear motor drive

No Contact based Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.



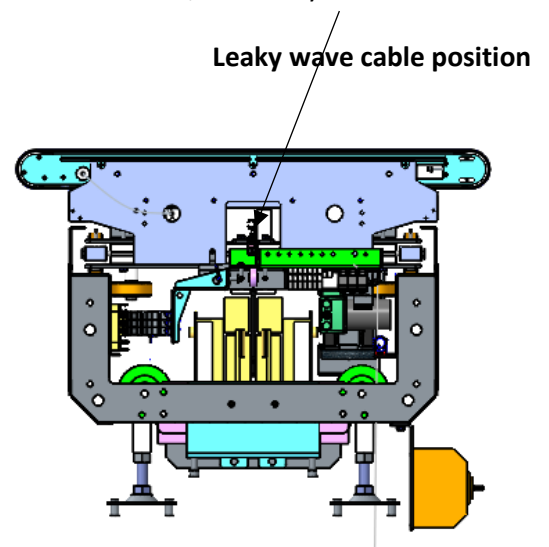
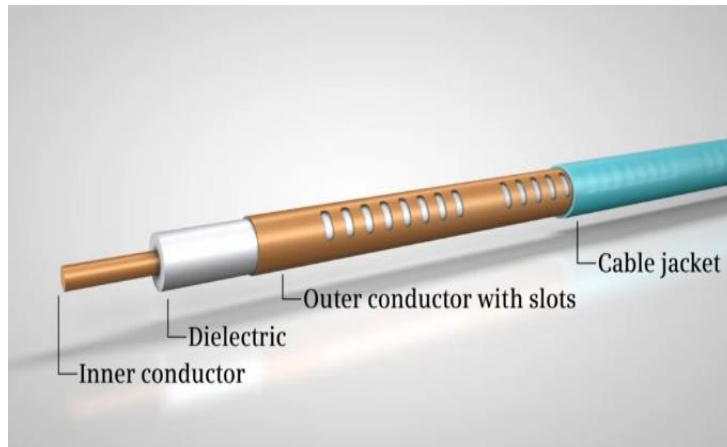
9.2.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



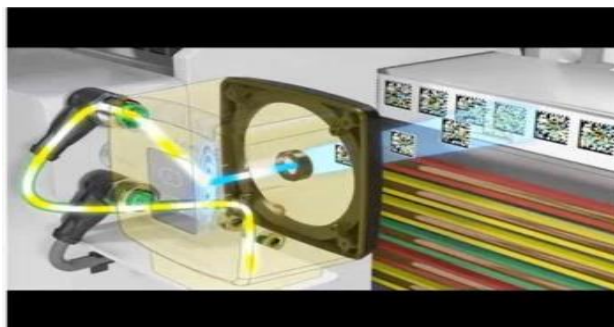
9.2.7 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



9.2.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



9.2.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2 m/s
Sorter Loop Length Per Deck	~135 m
Sorter Actuation Technology	Electric
Chutes	
PTL Chutes	45
Rejection Chute	3
Irregular Chute	6
Track Material to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	115
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during Detailed Design
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos per Zone
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

9.3 Scanning & Sensing on Main Sorter Loop

9.3.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.



9.3.2 Parcel Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



9.3.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the parcels on the empty carriers. (Camera based)

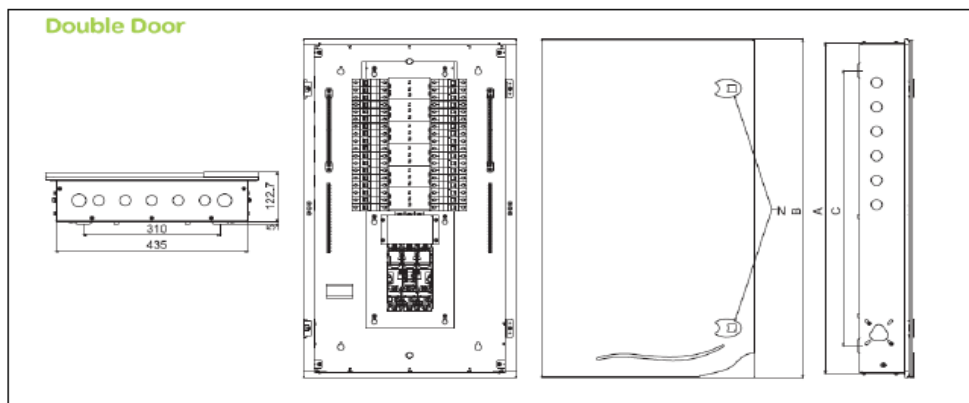


10. Electrical System

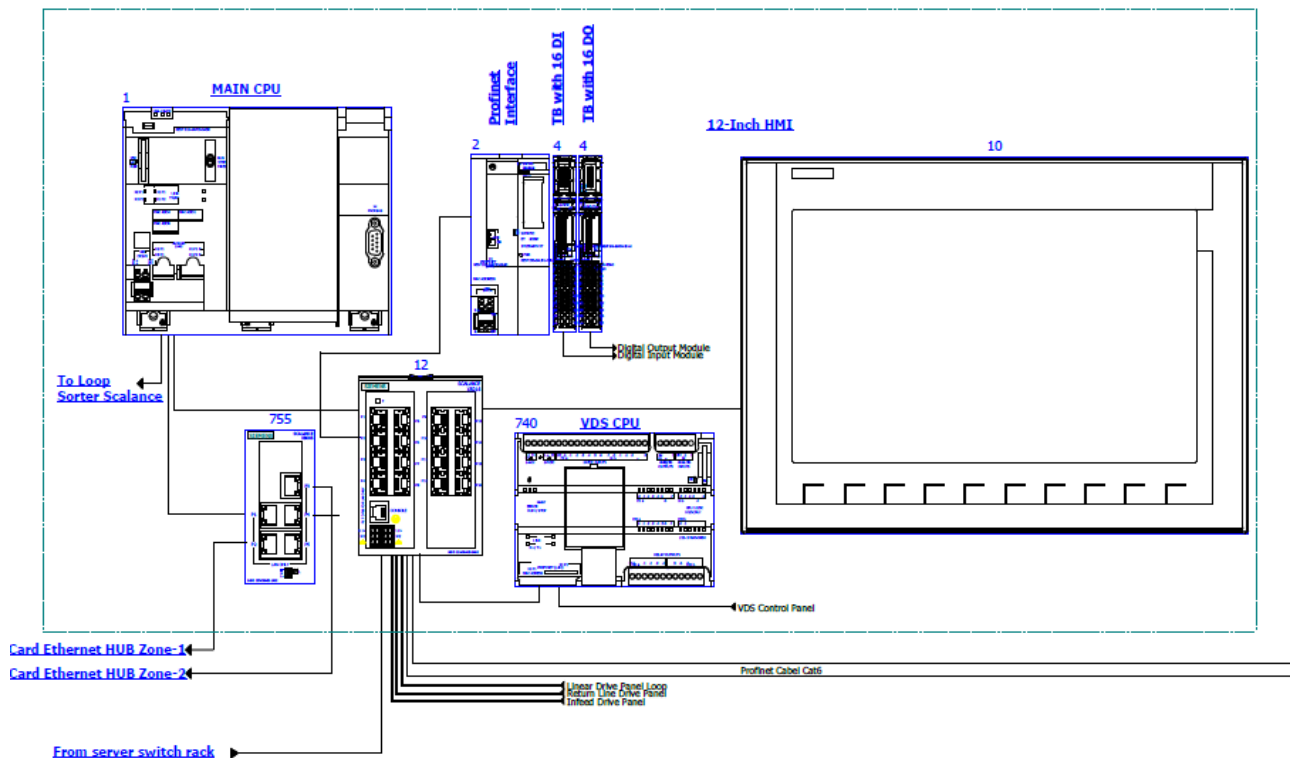
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets. PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

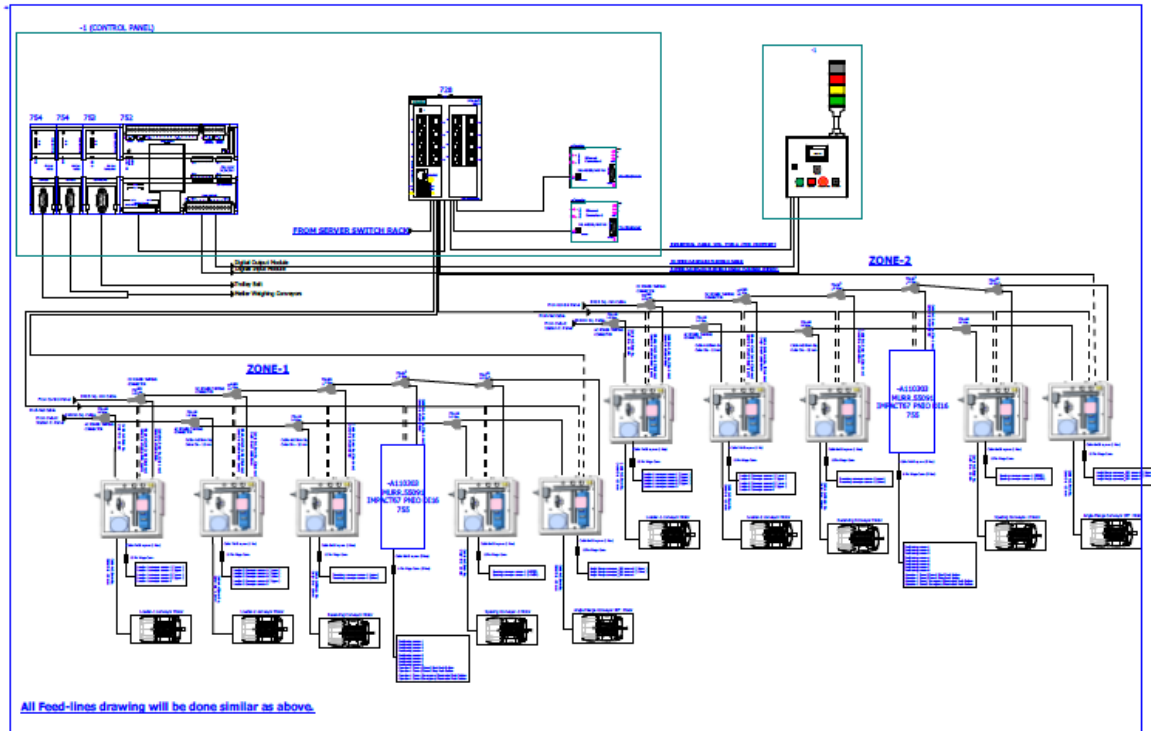
10.1.1 Reference Picture of Power Distribution Panel



10.1.2 Main Control Panel (Reference)



10.1.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter. The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.

All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

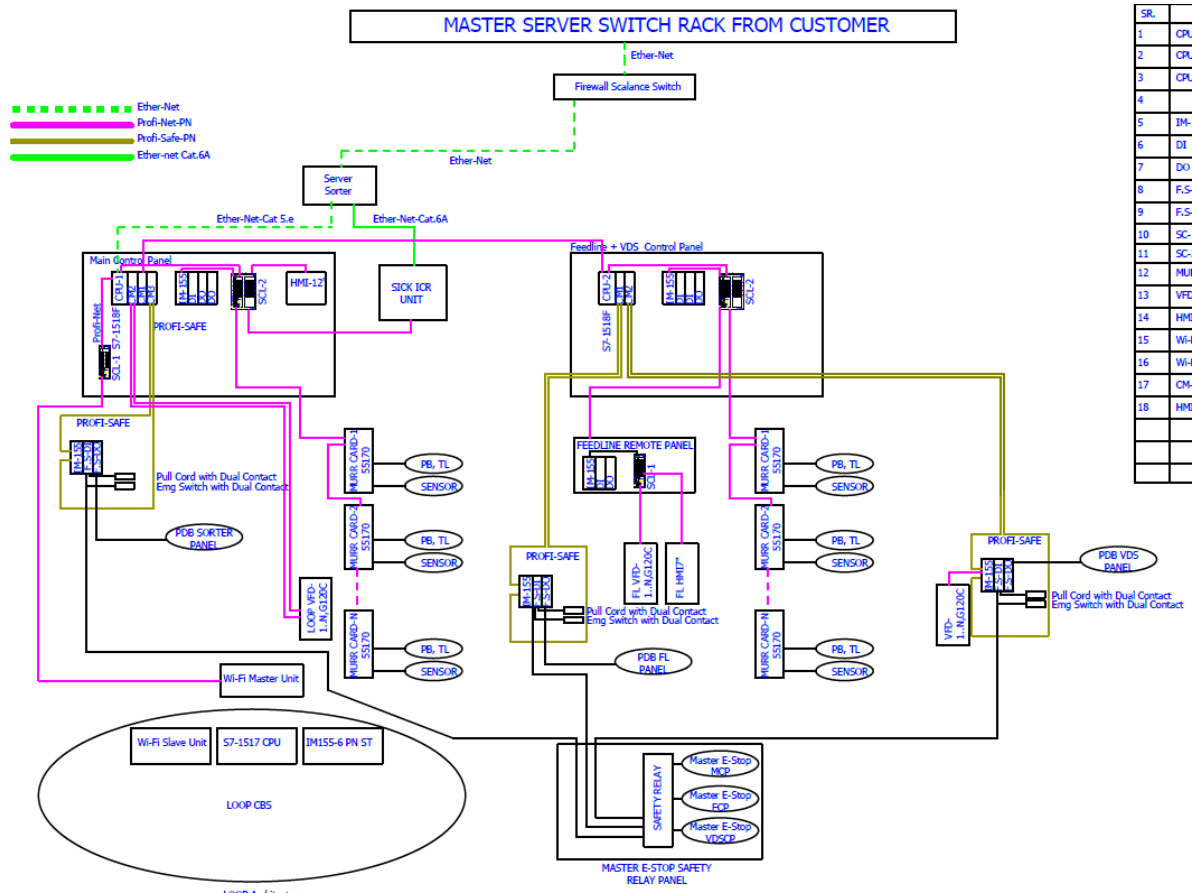
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

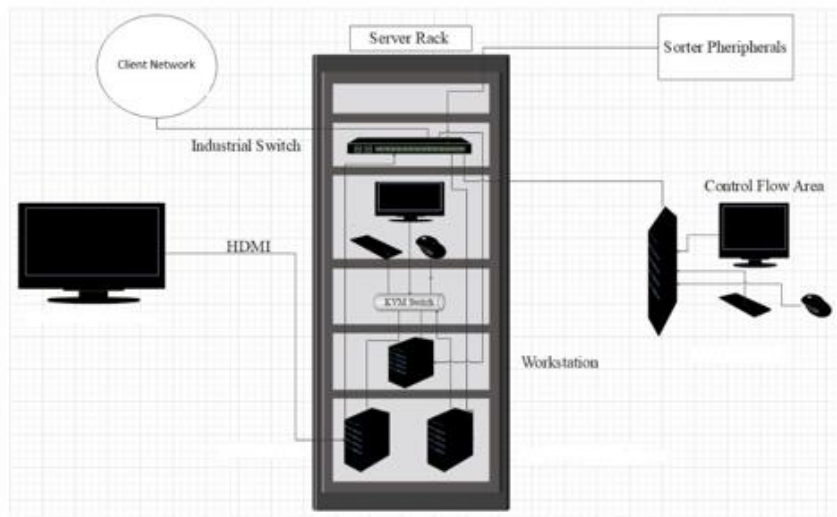
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

11. Controls Architecture (Ref.)



12. IT Architecture (Ref.)



13. Key Components Make

Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Volume Scanners	SICK/Cognex
Encoders	SICK/Falcon
Sensors	Sick/Leuze/ P&F
PLC	Siemens's
Control Panels	Rittal/ BCH
Weighing Scales	Bizerba/Mettler/Equivalent
VFDs	Siemens/Lenze/Omron
Cables	LAPP Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF/Equivalent
Data Transmission System	Siemens
Power Transmission Systems	Vahle
HMI's	Siemens/Omron

14. Program Organisation

14.1 Program/ Project Schedule

System will be ready for UAT by 31st Oct 2024 .

**, detailed plan will be shared during DAP.*

14.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with DELHIVERY
3. Weekly/Fortnightly Meeting DELHIVERY to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

15. DELHIVERY'S Responsibility

15.1 DELHIVERY Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the

operation of the parcelsorting system free of charge at the date of FALCON need.

- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

15.2 Responsibilities of DELHIVERY During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the parcels correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

15.3 DELHIVERY Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of DELHIVERY.

16. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (**Starts from the date of beneficiary use**) and post completion of warranty, customer can opt for Extended warranty.

Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

17. Commercial Terms

17.1 Price Sheet

S. No	Component	Qty per set	Set	UOM	Total (INR)
1	Cross Belt Sorter Package	1	1	Set	₹ 4,78,05,779
2	Conveyor Package	1	1	Set	₹ 1,72,01,379
3	Chute Package	1	1	Set	₹ 51,12,716
4	PTL Package	1	1	Set	₹ 68,69,920
6	Toolkit	1	1	Set	₹ 79,800
7	Software Packages	1	1	Set	₹ 6,00,000
8	Installation & Commissioning	1	1	Set	₹ 38,53,480
9	Packaging & forwarding	1	1	Set	₹ 15,41,392
10	Project Management and Engineering Charges	1	1	Set	₹ 4,15,322
Total					₹ 8,34,79,788
BCA					4.50%
Final Price					₹ 7,97,23,198

The Total Price is to be understood.

- Freight- Excluded
- Taxes: Extra as Applicable
- Including Packing and Forwarding
- Installation and Commissioning: Included
- Price is valid for 30 days from the date of proposal.

*Material Unloading & Laydown area are in customer scope

17.2 Payment Terms

S.No	Percentage	Stage
1	40%	Along with Purchase Order within 7 days of contract signing
2	50%	Against Delivery of material at site with 100% Taxes on Pro Data Basis
3	10%	Against Installation & Commissioning

**All invoices (if due as per payment terms) should be cleared within 30 days from the date of receipt of undisputed invoice by the buyer*

18. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Construction Power**
- **Mezannine**
- **Steel Works- If not specified**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.